

Order Megaloptera

Dobsonflies and alderflies

These are the most primitive insects to exhibit metamorphosis. Three hundred species in 2 families are weak fliers and never far from water. Larvae are aggressive, aquatic predators. They pupate in soil just above water level. Once metamorphosed, the adults never feed.

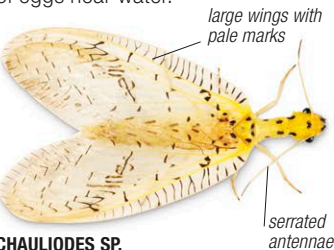
Family Corydalidae

Dobsonflies

Occurrence 200 spp. worldwide, especially in temperate regions; in running water



Measuring 1–3 in (5–7.5 cm) in length, with a wingspan of up to 6 in (15 cm), dobsonflies are large, soft-bodied, gray or brown insects, with 3 ocelli. Males may have large mandibles, used not for feeding but for combat or for holding the female. The latter lays masses of hundreds of thousands of eggs near water.



CHAULIODES SP.
Members of this North American genus called fish flies have distinctly rounded corners on their heads.

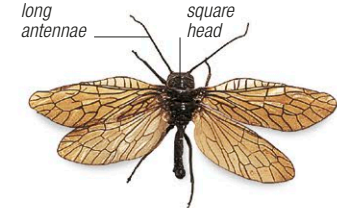
Family Sialidae

Alderflies

Occurrence 100 spp. worldwide, especially in temperate regions; in muddy ponds, canals, slow-moving water



Alderflies are dark and smoky-winged; most species are less than 1/2 in (1.5 cm) long. They have long, threadlike antennae. Unlike dobsonflies they have no ocelli. Females lay egg masses near water, from which hatched larvae drop. These have fully formed legs for crawling, and often mature in a year.



SIALIS SP.
Common in the Northern Hemisphere, insects of this genus have typically smoky wings with dark veins. Fully grown larvae crawl out of water and pupate in damp soil above the water margin.

Order Raphidioptera

Snakeflies

Snakeflies get their name from the snake-like way in which they catch their prey—raising their head and seizing the prey with a sudden lunge. There are 2 families and about 200 species. Metamorphosis is complete. Larvae hide in bark and litter, and are equally voracious.

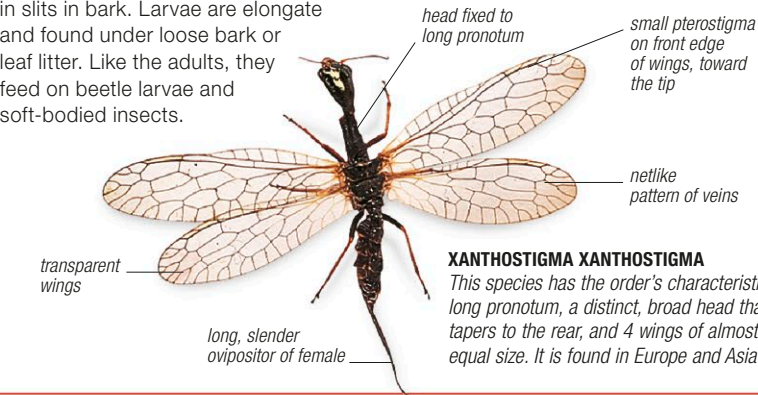
Family Raphidiidae

Snakeflies

Occurrence 185 spp. primarily in the Northern Hemisphere; among vegetation



Snakeflies are dark, with a distinct neck formed by a long pronotum. Body length varies between 1/4–1 1/4 in (0.6 and 2.8 cm). The head is broad in the front and tapered at the rear. Females, slightly larger than males, use their long ovipositor to lay eggs in slits in bark. Larvae are elongate and found under loose bark or leaf litter. Like the adults, they feed on beetle larvae and soft-bodied insects.



AGULLA SP.
These snakeflies are found in the woodland of the USA and Canada, from the Rocky Mountains to the Pacific coast.

XANTHOSTIGMA XANTHOSTIGMA
This species has the order's characteristic long pronotum, a distinct, broad head that tapers to the rear, and 4 wings of almost equal size. It is found in Europe and Asia.

Order Neuroptera

Antlions, lacewings, and others

This order comprises 14 families and about 11,000 species, characterized by large eyes, chewing mouthparts, long antennae, and 2 pairs of filmy, net-veined wings held over the body when at rest. Metamorphosis is complete. The adults are generally predatory and nocturnal, while the larvae are usually predatory or parasitic.

Family Chrysopidae

Common lacewings

Occurrence 2,500 spp. worldwide; in a range of microhabitats including vegetation, ant nests



Common lacewings are green or brown, with iridescent wings that span 3/8–2 in (1–5 cm) and carry a complex pattern of veins. The wings also have ultrasound sensors to detect predators such as bats. Adults are nocturnal and prey upon aphids, thrips, and mites.



NOTHOCHRYSA CAPITATA
Like other lacewings, this European species lays eggs on long stalks. The stalks help protect the eggs from attack by ants and other predators.

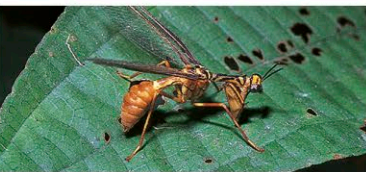
Family Mantispidae

Mantispids

Occurrence 600 spp. worldwide, mainly in warm temperate to tropical regions; on plants in well-vegetated areas



The front legs of mantispids resemble those of praying mantids (see p.554), and are used in the same way. Their wingspan is 3/8–2 1/4 in (1–5.5 cm). Eggs are laid on tree bark, and the larvae of most species feed on spider eggs.



CLIMACIELLA SP.
Members of this Central and South American genus have bright, wasplike, protective coloration.

Family Myrmeleontidae

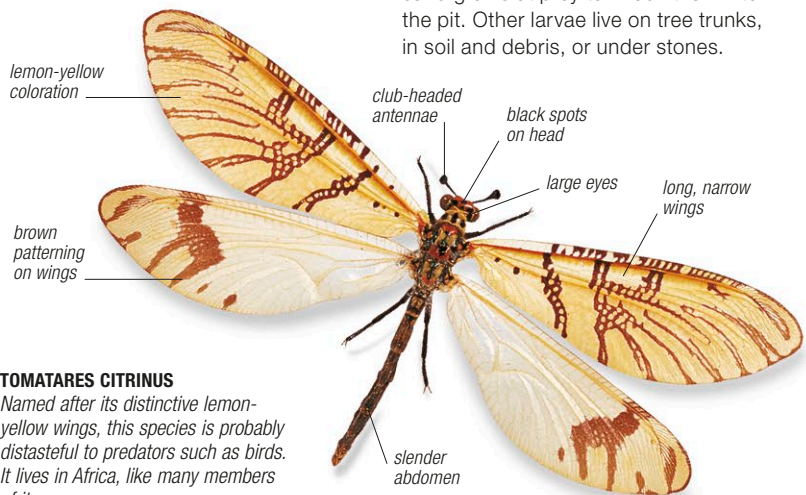
Antlions

Occurrence 2,000 spp. worldwide, in semiarid areas in subtropical and tropical regions; in woodland, scrub



Antlions are soft, large, and slender insects that resemble damselflies (see p.551). The long, narrow wings, which span 1 1/2–4 3/4 in (3.5–12 cm), may have brown or black patterns and produce a

weak, fluttering flight. The head is broader than the pronotum, with large, conspicuous eyes. The club-headed antennae are about as long as the head and thorax together. The abdomen in males has terminal sexual claspings organs that look like earwigs' forceps. Eggs are laid in soil or sand, singly or in small groups. Adults and larvae catch and eat insects and spiders; some species construct conical pits to trap prey. The larvae, sometimes called doodlebugs, live in these pits with only their spiny mandibles showing, and flick sand grains at prey to knock them into the pit. Other larvae live on tree trunks, in soil and debris, or under stones.



TOMATARES CITRINUS
Named after its distinctive lemon-yellow wings, this species is probably distasteful to predators such as birds. It lives in Africa, like many members of its genus.